20 Progressive C++ Array Tasks

Practical ideas organized from beginner foundations to two-dimensional array challenges

1	**Level 1 - Foundations**	Declaration, indexing, input, updating, and traversal
2	**Level 2 - Calculations and Search**	Accumulation, conditions, comparisons, and linear search
3	**Level 3 - Transformations**	Frequency, ordering, uniqueness, and rearrangement
4	**Level 4 - 2D Array Challenges**	Matrices, nested loops, diagonals, and practical data tables

Suggested rule: solve the tasks with ordinary C++ arrays and loops. Use vector, map, set, and built-in sorting only when the instructor permits them.

Level 1 - Foundations

Declaration, indexing, input, updating, and traversal

1

Weekly temperature tracker

Store the temperature for seven days. Print the complete week, then let the user update the value for one selected day.

Focus: Declaration, indexing, update

2

First, middle, and last

Read seven integers and print the first element, the middle element, and the last element using their indexes.

Focus: Input and element access

3

Reverse display

Store ten numbers and print them from the last element to the first without creating another array.

Focus: Reverse traversal

4

Transform every value

Read six integers, multiply every element by two, and print the array before and after the transformation.

Focus: Traversal and modification

5

Simple inventory update

Store the quantities of eight products. Ask for a product index and a new quantity, then display the updated inventory.

Focus: Index validation and update

Level 2 - Calculations and Search

Accumulation, conditions, comparisons, and linear search

6

Sum and average

Read ten values, calculate their sum and average, and print both results. Use a floating-point type for the average.

Focus: Accumulator and division

7

Maximum and minimum

Find the largest and smallest values in an array. Print each value together with the index where it first appears.

Focus: Comparison and index tracking

8

Even and odd counter

Count the even and odd elements in an integer array, then report which group contains more values.

Focus: Conditions and counters

9

Search all occurrences

Ask for a target value and print every index where it appears. Print a clear message when the target is absent.

Focus: Linear search

10

Values above average

Calculate the array average, then print the values above the average and count how many there are.

Focus: Two-pass traversal

Level 3 - Transformations

Frequency, ordering, uniqueness, and rearrangement

11

Second-largest distinct value

Find the second-largest distinct element without sorting the array. Handle the case where a second distinct value does not exist.

Focus: State tracking

12

Frequency table

For every distinct value in a small integer array, print how many times it occurs. Do not use map, set, or vector.

Focus: Nested loops and counting

13

Unique values only

Print every value once, preserving the order of its first appearance. Use ordinary arrays to track processed values.

Focus: Duplicate detection

14

Rotate right

Rotate an array one position to the right. Extend the program so the user can request several rotations.

Focus: Shifting and temporary storage

15

Sort and merge

Read two arrays, sort each one, and merge them into a single sorted array without using a built-in sorting function.

Focus: Sorting and merging

Level 4 - 2D Array Challenges

Matrices, nested loops, diagonals, and practical data tables

16

Row and column reports

For a 3 x 4 matrix, calculate every row total and every column total. Identify the row with the greatest total.

Focus: Nested loops and aggregation

17

Diagonal analysis

For a square matrix, calculate the main diagonal and secondary diagonal sums. Determine which diagonal has the larger sum.

Focus: Index relationships

18

Matrix transpose

Create the transpose of a matrix by converting its rows into columns. Print both the original and transposed matrices.

Focus: 2D indexing

19

Matrix addition

Read two matrices with the same dimensions, add their corresponding elements, and store the result in a third matrix.

Focus: Parallel 2D traversal

20

Student grade dashboard

Store multiple students and exam scores in a 2D array. Print each student's total and average, each exam average, the highest score, and the student with the highest average.

Focus: Integrated array problem